



SCEGGS Darlinghurst

2010

**Preliminary Course
Semester 2 Examination**

Mathematics

**Outcomes Assessed: P2 – P8
Task Weighting: 40%**

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using blue or black pen
- Attempt **all** questions and show all necessary working
- Answer all questions on the pad paper provided
- Write your Student Number at the top of each page
- **Begin each question on a new page**
- Marks will be deducted for careless or badly arranged work
- Mathematical templates, geometrical equipment and scientific calculators may be used

--	--	--	--	--

Centre Number

--	--	--	--	--	--	--	--

Student Number

Total marks – 78

- Attempt Questions 1 – 6

Question	Calc	Comm	Reasoning	Marks
1				/13
2				/13
3				/13
4				/13
5				/13
6				/13
TOTAL				/78

BLANK PAGE

Question 1 (13 marks)

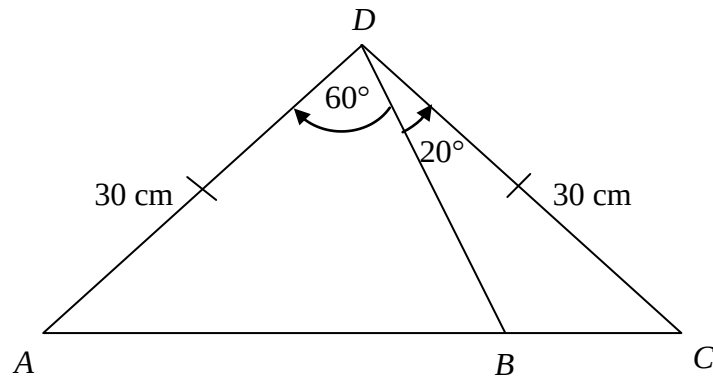
- (a) Express $0.8\dot{3}$ in the form $\frac{a}{b}$, where a and b are positive integers. **2**
- (b) The crowd attendance at a football game on the weekend was 15 660 which was a decline of 10% from the previous weekend. How many people attended the previous weekend? **2**
- (c) Factorise fully: $3x^2 - 7x - 6$. **2**
- (d) Find the exact value of $\cos 225^\circ$. **2**
- (e) Find the value of v in $v^2 = u^2 - 2as$ when $u = 6.4 \times 10^7$, $a = 21.5$ and $s = 5.7 \times 10^{13}$. Give your answer to 3 significant figures. **2**
- (f) Find the values of a and b if $\frac{3}{\sqrt{5}-2} = \sqrt{a} + b$. **3**

	Marks
Question 2 (13 marks)	
(a) Find the size of each interior angle of a regular nonagon (9-sided polygon).	2
(b) Solve each of the following:	
(i) $ 2x - 5 \leq 7$	2
(ii) $3^{2x} - 10 \times 3^x + 9 = 0$	3
(c) Solve $\tan(\theta + 60^\circ) = \frac{1}{\sqrt{3}}$ for $0^\circ \leq \theta \leq 360^\circ$.	2
(d) α and β are the roots of the quadratic equation $x^2 - 2x + 3 = 0$. Find the values of:	
(i) $\alpha + \beta$	1
(ii) $\alpha^2 \beta^2$	1
(iii) $\alpha^{-2} + \beta^{-2}$	2

	Marks
Question 3 (13 marks)	
(a) Differentiate each of the following with respect to x :	
(i) $5x^3 + \frac{2}{x} - 3$	2
(ii) $x(2x + 3)^9$	2
(b) Find the equation of the tangent to the curve $y = \frac{x}{x + 2}$ at the point where $x = -1$. Give your answer in general form.	4
(c) Consider the quadratic function $y = kx^2 + 6x + k$.	
(i) Show that the function is positive definite when $k > 3$.	3
(ii) Sketch the graph of $y = 4x^2 + 6x + 4$ clearly showing any intercepts with the axes and the co-ordinates of the vertex.	2

Question 4 (13 marks)

- (a) In the diagram, AD and DC are equal to 30 cm.



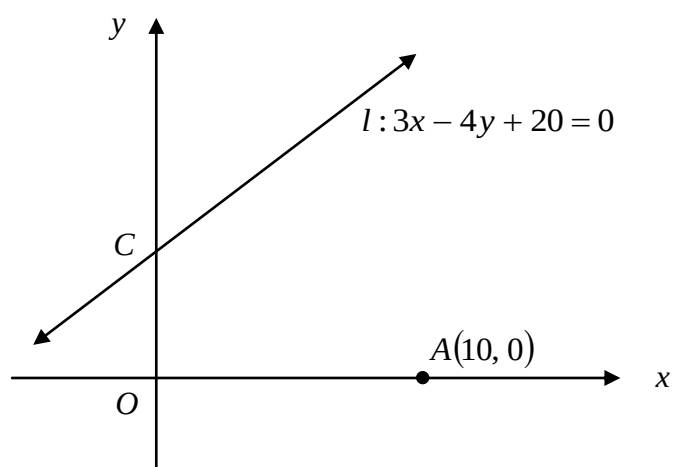
**NOT
TO
SCALE**

- | | | |
|------|--|---|
| (i) | Find the size of $\angle DBA$ giving reasons. | 2 |
| (ii) | Hence find the length of AB to the nearest centimetre. | 2 |

Question 4 continues on the next page

Question 4 (continued)

(b)



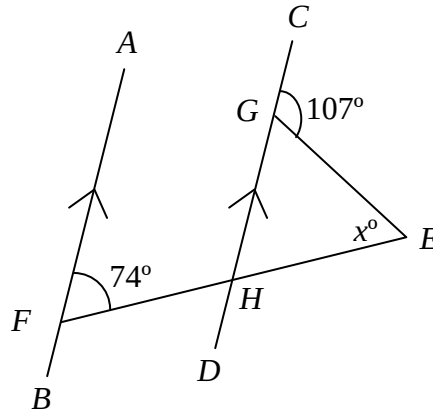
In the diagram A is the point $(10, 0)$ and line l has the equation $3x - 4y + 20 = 0$.

- | | | |
|-------|---|---|
| (i) | Find the co-ordinates of the point C where l intersects the y axis. | 1 |
| (ii) | B is on the line l such that AB is perpendicular to l .
Find the distance AB . | 2 |
| (iii) | Prove that $\triangle AOC \equiv \triangle ABC$. | 2 |
| (iv) | Find the area of the quadrilateral $OABC$. | 1 |
| (v) | Find the gradient of AB and hence find the angle which AB makes with the positive x axis. Give your answer to the nearest degree. | 3 |

Question 5 (13 marks)

(a)

2



Find the value of x giving reasons.

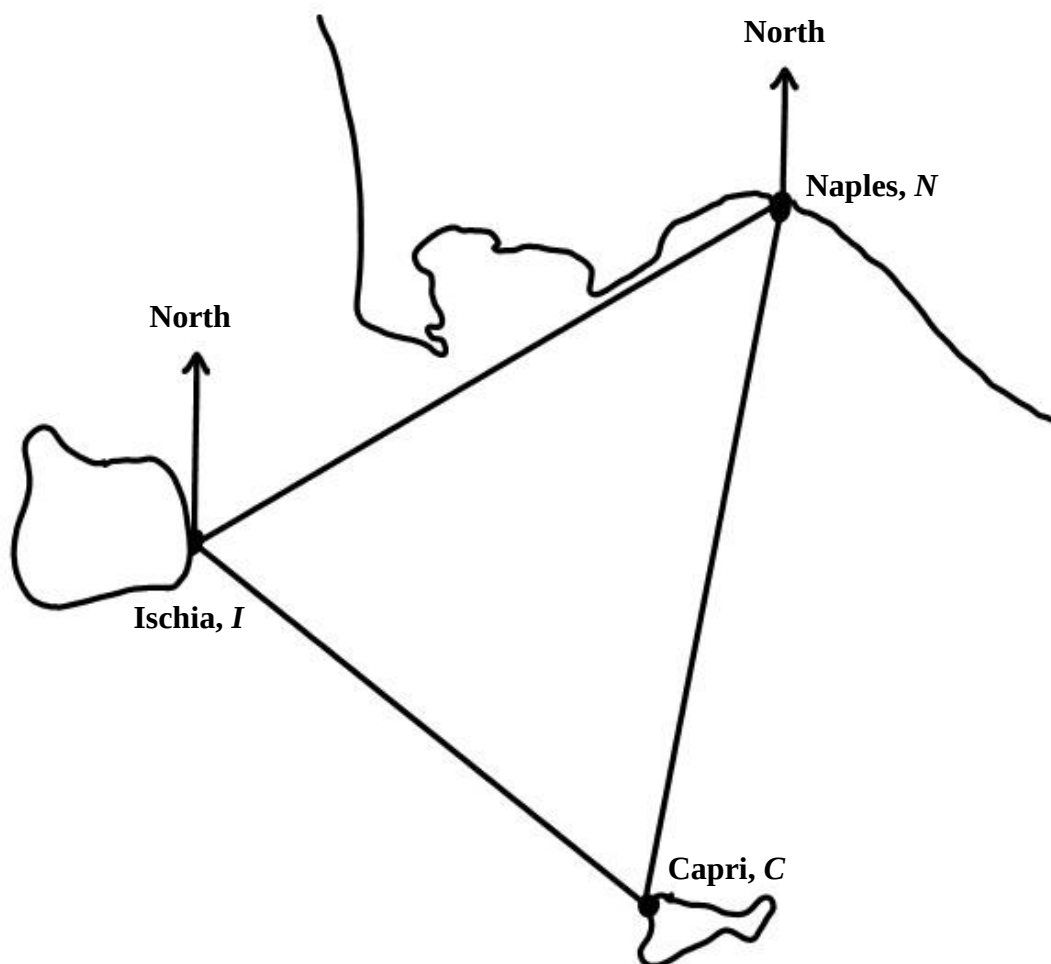
(b) Find a , b and c such that $x^2 - x + 3 \equiv a(x+1)^2 + b(x+1) + c$.

3

Question 5 continues on the next page

Question 5 (continued)

- (c) A boat leaves the Island of Ischia and travels 40 km on a bearing of $060^\circ T$ to Naples. It then turns to travel a further 45 km to the Island of Capri on a bearing of $200^\circ T$.

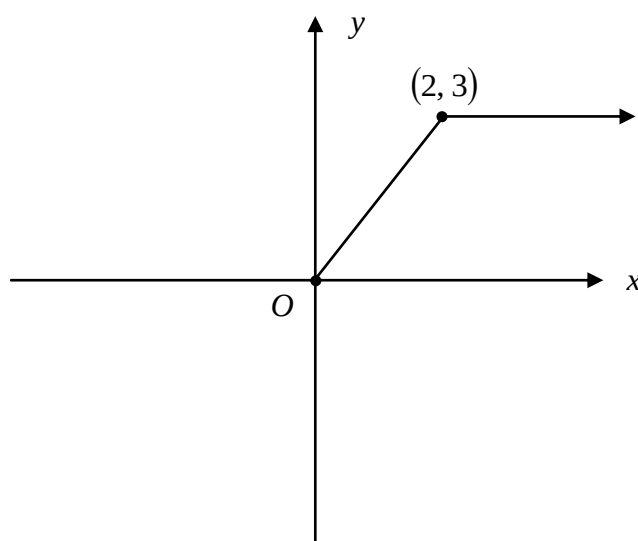


- | | | |
|-------|--|---|
| (i) | Find the size of $\angle INC$. | 1 |
| (ii) | Find the distance between the islands of Ischia and Capri to the nearest kilometre. | 2 |
| (iii) | Find the area of $\triangle INC$. Give your answer in square kilometres to 2 significant figures. | 2 |

Question 5 continues on the next page

Question 5 (continued)

(d)



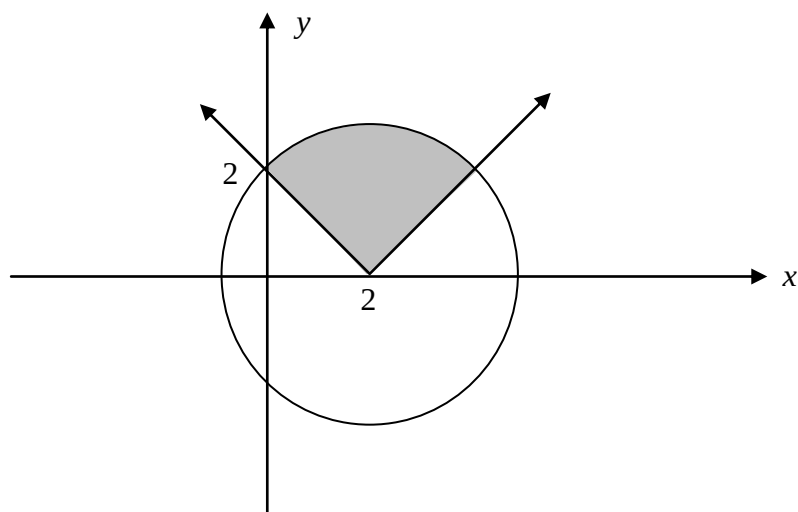
- | | | |
|------|--|----------|
| (i) | Copy and complete the sketch so that it is an odd function. | 1 |
| (ii) | Write down the equation of the piecewise function drawn in part (i). | 2 |

Question 6 (13 marks)

(a) (i) Write down the factorised form of $a^3 - b^3$. 1

(ii) Hence show that $\frac{\sin^3 \theta - \cos^3 \theta}{\sin \theta - \cos \theta} - 1 = \sin \theta \cos \theta$. 2

(b)



Write down a pair of inequations to describe the region shaded above. 4

(c) $f(x) = \frac{1}{\sqrt{x}}$

(i) State the domain and range of $f(x)$. 2

(ii) For what value of x does $f(x) = 2$? 1

(iii) Find $f'(x)$ and hence calculate $f'(3)$ leaving your answer in simplified surd form with a rational denominator. 2

(iv) Show that $(f(x))^3 = -2f'(x)$. 1

End of Paper

Question 1

(a) let $x = 0.8333\dots$

$$10x = 8.3333\dots$$

$$\Rightarrow 9x = 7.5 \quad \checkmark$$

$$90x = 75$$

$$x = \frac{75}{90} = \frac{5}{6} \quad \checkmark$$

(b) $90\% = 15660$ people $\checkmark$

$$10\% = 1740 \text{ people}$$

$$100\% = 17400 \text{ people}$$

$\therefore 17400$ attended the previous weekend $\checkmark$

(c) $3x^2 - 7x - 6$

$$= \frac{(3x-9)(3x+2)}{3} \quad \checkmark$$

$$= (x-3)(3x+2) \quad \checkmark$$

one mark if signs incorrect.

(d) $\cos 225^\circ = -\cos 45^\circ$

$$= -\frac{1}{\sqrt{2}} \quad \checkmark$$



(e) $v^2 = (6.4 \times 10^7)^2 - 2 \times 21.5 \times 5.7 \times 10^{13}$

$$= 1.645 \times 10^{15} \quad \checkmark$$

$$v = \sqrt{1.645 \times 10^{15}}$$

$$= 40558599.58 \quad \checkmark$$

$$\doteq 40600000 \quad (3 \text{ sig fig})$$

Many students stopped at this point!

$$\begin{aligned} \text{(f)} \quad \frac{3}{\sqrt{5}-2} \times \frac{\sqrt{5}+2}{\sqrt{5}+2} &= \frac{3\sqrt{5}+6}{5-4} \quad \checkmark \\ &= \sqrt{45}+6 \quad \checkmark \end{aligned}$$

$$\therefore a = 45, b = 6$$

many students rounded this to 406!!

Question 2

(a) Angle sum = $180 \times (n-2)$
Angle sum of 9 sided shape = $180 \times (9-2)$
 $= 1260^\circ$ ✓

Each angle = $1260^\circ \div 9$
 $= 140^\circ$ ✓

(b) (i) $|2x-5| \leq 7$
 $-7 \leq 2x-5 \leq 7$
 $-2 \leq 2x \leq 12$
 $-1 \leq x \leq 6$ ✓✓

(ii) $3^{2x} - 10 \times 3^x + 9 = 0$
let $u = 3^x$
 $u^2 - 10u + 9 = 0$
 $(u-9)(u-1) = 0$
 $u = 9, u = 1$ ✓
 $3^x = 9, 3^x = 1$
 $x = 2, x = 0$ ✓

(c) $\tan(\theta + 60^\circ) = \frac{1}{\sqrt{3}}$ rel. L = 30°
 $\begin{array}{c|c} S & A \\ \hline T & C \end{array}$
 $\theta + 60^\circ = 30^\circ, 210^\circ, 390^\circ, \dots$ ✓
 $\theta = -30^\circ, 150^\circ, 330^\circ, \dots$
 $\theta = 150^\circ, 330^\circ$ ✓ for $0 \leq \theta < 360^\circ$

(d) (i) $\alpha + \beta = \frac{-b}{a} = \frac{-2}{1} = 2$ ✓
(ii) $\alpha\beta = \frac{c}{a} = \frac{3}{1} = 3$
 $\alpha^2\beta^2 = 9$ ✓
(iii) $\alpha^{-2} + \beta^{-2} = \frac{1}{\alpha^2} + \frac{1}{\beta^2}$
 $= \frac{\beta^2 + \alpha^2}{\alpha^2\beta^2}$
 $= \frac{(\alpha + \beta)^2 - 2\alpha\beta}{\alpha^2\beta^2}$ ✓
 $= \frac{2^2 - 2 \times 3}{9} = \frac{-2}{9}$ ✓

Well done.

Be careful to rewrite solution

Learn this method
Those students that learnt method were awarded full marks.

Reas 3

Poorly done by most students
you needed to -60 from your solution, not add to find θ .

Well done (i) + (ii)

$$\alpha^2\beta^2 \neq \alpha^2 + \beta^2$$

Index Laws required

$$x^{-2} = \frac{1}{x^2}$$

Reas 2

Question 3

(a) (i) $15x^2 - 2x^{-2}$ ✓✓

(ii) $x \times 9(2x+3)^8 \times 2 + (2x+3)^9 \times 1$
 $= 18x(2x+3)^8 + (2x+3)^9$ ✓✓
($uv' + vu'$)

Calc 4.

(b) $y = \frac{x}{x+2}$

$$\frac{dy}{dx} = \frac{(x+2) \times 1 - x \times 1}{(x+2)^2} \quad \left(\frac{vu' - uv'}{v^2} \right)$$
$$= \frac{2}{(x+2)^2} \quad \checkmark$$

$$x = -1 \rightarrow m_T = \frac{2}{(-1+2)^2} = 2 \quad \checkmark$$

$$y = \frac{-1}{-1+2} = -1 \quad \checkmark$$

Equation of tangent :

$$y - (-1) = 2(x - (-1))$$

$$y + 1 = 2x + 2$$

$$2x - y + 1 = 0 \quad \checkmark$$

Calc 4

(c) (i) Positive definite $a > 0, \Delta < 0$.

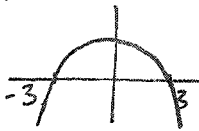
$$\Delta < 0$$

$$b^2 - 4 \times k \times k < 0$$

$$36 - 4k^2 < 0 \quad \checkmark$$

$$(6-2k)(6+2k) < 0$$

$$k < -3, \quad k > 3 \quad \checkmark$$



Since $a > 0$ for positive definite
 $k > 3$. ✓

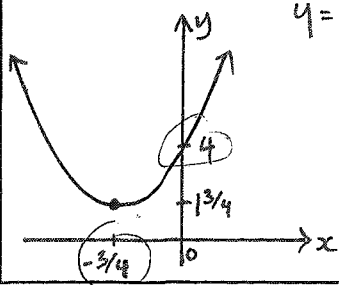
Reas 3.

(ii) y int $\rightarrow x = 0 \quad y = 4$ ✓

$$\text{Vertex} \rightarrow x = \frac{-b}{2a} = \frac{-6}{2 \times 4} = -\frac{3}{4}$$

$$y = 4\left(-\frac{3}{4}\right)^2 + 6\left(-\frac{3}{4}\right) + 4 = 1\frac{3}{4}$$

(both labelled on graph) ✓



Comm 2.

Question 4

(a) (i) $\angle DAC = \frac{180-80}{2} = 50^\circ$ ✓

(angles opposite equal sides of a triangle are equal and angle sum of triangle is 180°)

$\angle DBA = 180 - 60 - 50 = 70^\circ$ ✓

(angle sum of $\triangle DAB$ is 180°)

(ii) $\frac{AB}{\sin 60^\circ} = \frac{30}{\sin 70^\circ}$ ✓

$AB = \sin 60^\circ \times \frac{30}{\sin 70^\circ}$

$= 27.648 \dots$

$\approx 28 \text{ cm}$ ✓

(b) (i) C: $x=0$ in $3x-4y+20=0$

$3 \times 0 - 4y + 20 = 0$

$4y = 20$

$y = 5$

$\therefore C = (0, 5)$ ✓

(ii) $\perp d$ from $(10, 0)$ to $3x-4y+20=0$

$AB = \frac{|3 \times 10 - 4 \times 0 + 20|}{\sqrt{3^2 + (-4)^2}}$

$= 10 \text{ units}$ ✓✓

(iii) In $\triangle AOC$ & $\triangle ABC$

$\angle AOC = \angle ABC = 90^\circ$ (given $AB \perp l$ and axes are $\perp$)

AC is common

$AO = AB = 10$ (given $A = (10, 0)$ & $AB = 10$ part ii)

$\therefore \triangle AOC \equiv \triangle ABC$ (RHS) ✓✓

(iv) Area $= 2 \times \left(\frac{1}{2} \times 10 \times 5\right) = 50 \text{ u}^2$ ✓

(v) $M_l: 3x-4y+20=0$

$y = \frac{3}{4}x + 5$

gradient $= \frac{3}{4}$ ✓

$m_{AB} = -\frac{4}{3}$ ✓

Comm 2.

Reas 2.

$\rightarrow \tan \theta = -\frac{4}{3}$

related angle $= 53^\circ$

$\theta = 180 - 53$

$= 127^\circ$ ✓

Reas 3.

Question 5

(a) $\angle GHE = 74^\circ$ (corresponding angles are equal when lines are parallel) ✓
 $x^\circ = 107^\circ - 74^\circ = 33^\circ$ (exterior angle of a triangle is the sum of the two opposite interior angles) ✓

$$\begin{aligned} (b) \quad x^2 - x + 3 &\equiv a(x+1)^2 + b(x+1) + c \\ &\equiv a(x^2 + 2x + 1) + bx + b + c \\ &\equiv ax^2 + (2a+b)x + (a+b+c) \end{aligned}$$

Comparing coefficients:

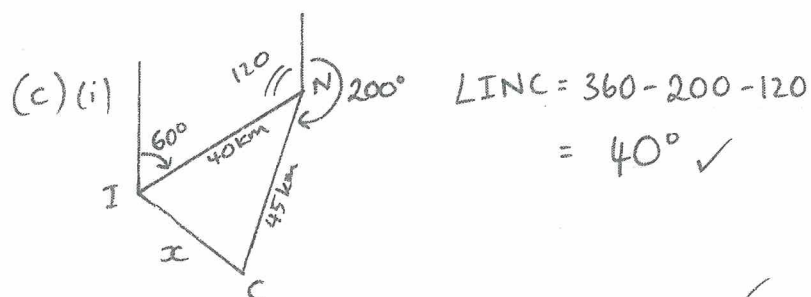
$$x^2: a = 1 \quad \checkmark$$

$$x: 2a + b = -1 \quad (\& a = 1)$$

$$\Rightarrow b = -3 \quad \checkmark$$

$$\text{constant: } a + b + c = 3 \quad (\& a = 1, b = -3)$$

$$\Rightarrow c = 5 \quad \checkmark$$



$$(ii) \quad x^2 = 40^2 + 45^2 - 2 \times 40 \times 45 \times \cos 40^\circ \checkmark$$

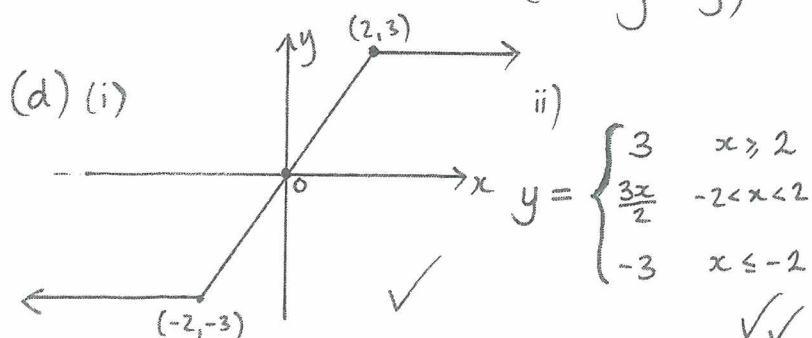
$$= 867.24$$

$$x = \sqrt{867.24} \doteq 29 \text{ km} \quad \checkmark$$

$$(iii) \quad \text{Area} = \frac{1}{2} \times 40 \times 45 \times \sin 40^\circ \checkmark$$

$$\doteq 578.5088 \dots$$

$$\doteq 580 \text{ km}^2 \quad (2 \text{ sig fig}) \checkmark$$



Comm 2.

Correct reason must have been given to be awarded 2 marks.

Learn this method.

careless errors made when substituting.

Advice is to redraw diagram + label information given.

Incorrect (i) did not result in penalties in part (ii) + (iii) as long as method was correct.

Must have 2 sig fig

Label graph (no marks deducted this time!)

Comm 1

Reas 2.

Piecewise function very poorly done.

Question 6

(a) (i) $a^3 - b^3 = (a-b)(a^2 + ab + b^2)$ ✓

(ii) LHS = $\frac{\sin^3 \theta - \cos^3 \theta}{\sin \theta - \cos \theta} - 1$

$$= \frac{(\sin \theta - \cos \theta)(\sin^2 \theta + \sin \theta \cos \theta + \cos^2 \theta)}{(\sin \theta - \cos \theta)} - 1 \quad \checkmark R$$

$$= \sin^2 \theta + \cos^2 \theta + \sin \theta \cos \theta - 1$$

$$= 1 + \sin \theta \cos \theta - 1 \quad \checkmark R$$

$$= \sin \theta \cos \theta = \text{RHS}$$

Reas 2.

(b) Absolute value: $y = |x - 2|$ ✓

Circle: radius = $\sqrt{2^2 + 2^2} = \sqrt{8}$ ✓

centre = $(2, 0)$

$$(x-2)^2 + y^2 = 8 \quad \checkmark R$$

Inequations: $y \geq |x - 2|$

and $(x-2)^2 + y^2 \leq 8$ ✓

Reas 4

(c) (i) D: $x > 0$ ✓ R: $y > 0$ ✓

(ii) $f(x) = 2$

$$\frac{1}{\sqrt{x}} = 2$$

$$1 = 2\sqrt{x}$$

$$\frac{1}{2} = \sqrt{x}$$

$$x = \frac{1}{4} \quad \checkmark$$

(iii) $f(x) = x^{-1/2}$

$$f'(x) = -\frac{1}{2} x^{-3/2} \quad \checkmark C$$

$$= \frac{-1}{2x\sqrt{x}}$$

$$f'(3) = \frac{-1}{2 \times 3\sqrt{3}}$$

$$= \frac{-\sqrt{3}}{18} \quad \checkmark C$$

- many errors at this line.

Calc 2

$$\text{LHS} = (f(x))^3$$

$$= \left(\frac{1}{\sqrt{x}}\right)^3$$

$$= \frac{1}{x\sqrt{x}}$$

∴ LHS = RHS

$$\text{RHS} = -2 \times f'(x)$$

$$= -2 \times \frac{-1}{2x\sqrt{x}}$$

$$= \frac{1}{x\sqrt{x}} \quad \checkmark R$$

Reas 1

Setting out often very poor.